

705 A Research Background

706 The formal definition of Point-wise Mutual Information (PMI_k),
707 where $k > 2$, as proposed by Levine et al. [14].

$$\text{PMI}_k(w_1 \dots w_k) = \min_{\sigma \in \text{seg}(w_1 \dots w_k)} \log \frac{p(w_1 \dots w_k)}{\prod_{s \in \sigma} p(s)} \quad (1)$$

708 Here, $\text{seg}(w_1 \dots w_k)$ represents the set of all continuous segmen-
709 tations of the k -mer " $w_1 \dots w_k$ " (identity segmentation is not in-
710 cluded).

711 The formal definition of Normalized Point-wise Mutual Informa-
712 tion (NPMI_k), where $k > 2$, as proposed by Roy et al. [22].

$$\text{NPMI}_k(w_1 \dots w_k) = \text{PMI}_k * \frac{\log f(w_1 \dots w_k)}{\log(c) + \log f(w_1 \dots w_k)} \quad (2)$$

713 $f(w_1 \dots w_k)$ represents the number of times the k -mer sequence of
714 $w_1 \dots w_k$ occurs. The minimum frequency of occurrence is referred
715 to as c , which is used to filter rare tokens.

716 B Experimental Setup

717 B.1 Dataset Construction

718 The silencer dataset is obtained from the ‘Home Sapiens’ annotation
719 file³ from the *SilencerDB* database.

720 B.2 Training Details

721 **Finetuning parameter configuration.** The models are finetuned for
722 20 epochs at a learning rate of $4e^{-4}$, warmup steps percentage of
723 10%, hidden dropout probability of 0.1, weight decay as 0.01, per
724 GPU train batch size as 5 and use the *AdamW* optimizer for the 10,
725 50 and 100-shot setting. However, the performance drops in 500 and
726 1000-shot settings due to overfitting. Therefore, for the 500-shot set-
727 tings and above, we use the same hyperparameters as the original
728 DNABert paper — a lower learning rate from $4e^{-4}$ to $5e^{-5}$ and a
729 lower number of epochs from 20 to 5. Mosbach et al. [17] observe
730 that such a high number of fine-tuning epochs helps address random
731 initialization issues in the low-resource settings (10, 50-shot).

Parameter	DNABert	LOGO
Hidden Size	768	256
Hidden Layers	12	2
Attention Heads	12	8
Per GPU train batch size	10	5
Hidden Dropout Probability	0.1	0
Attention Dropout Probability	0.1	0
Intermediate Size	3072	3072
Embedding Size	512	512

Table 5. Difference between parameters of DNABert and LOGO

732 C Experimental Results

³ http://health.tsinghua.edu.cn/SilencerDB/download/Species/Homo_sapiens.fa

Task	Dataset	Models									
		State-of-art-models				DNABert			LOGO		
		DNABert-2 120K (BPE)★	DNABert 120K◇	NT-500M-Human◇	DNABert-2 120K (kmer)★	GeneMask 1K	CM-Step 1K	CM-GEMS 1K	GeneMask 1K	CM-Step 1K	CM-GEMS 1K
Species: Human											
PD	all	85.57	90.48	87.71	83.78	85.70	86.66	87.05	84.76	84.44	84.70
(Human)	no tata	92.55	93.05	90.75	92.65	89.59	89.50	89.14	89.32	89.23	89.30
	tata	60.85	61.56	78.07	57.75	64.81	69.06	68.46	64.43	70.62	64.41
CPD	all	66.28	68.90	63.45	74.91	65.93	65.98	66.02	62.77	63.72	63.43
(Human)	notata	67.99	70.47	64.82	69.23	67.00	67.03	66.91	66.31	67.52	66.80
	tata	72.73	76.06	71.34	74.91	69.74	64.35	74.10	66.08	61.53	70.74
TFP	0	66.99	66.84	61.59	67.99	66.33	65.34	63.03	63.56	61.52	62.97
(Human)	1	70.98	70.14	66.75	67.06	70.11	69.79	71.04	69.89	69.39	69.11
	2	61.40	61.03	53.58	59.45	63.35	61.68	61.97	53.27	53.24	55.88
	3	55.10	51.89	42.95	50.24	48.80	47.42	46.04	40.66	41.09	38.57
	4	71.31	70.97	60.81	72.80	75.63	73.03	75.32	70.29	72.68	72.80
Splice	Reconstruct	79.62	84.07	79.71	77.90	78.79	75.99	78.52	75.12	75.55	74.71
Mean (Human)		70.95	72.12	68.46	70.72	70.57	69.65	70.63	67.21	67.54	67.79
Species: non-Human											
EMP	H3	77.08	73.10	69.67	74.62	69.25	69.66	67.51	66.34	67.76	66.26
(Yeast)	H3K14ac	55.60	40.06	33.55	42.71	39.66	38.70	38.35	39.08	34.95	35.75
	H3K36me3	57.25	47.25	44.14	47.26	40.87	43.01	40.31	41.73	40.39	40.15
	H3K4me1	45.51	41.44	37.15	39.66	33.13	35.44	32.12	31.38	30.69	30.97
	H3K4me2	40.83	32.27	30.87	25.33	29.81	30.03	31.82	30.10	29.19	30.35
	H3K4me3	42.57	27.81	24.00	27.43	27.35	26.45	24.05	23.48	21.62	23.02
	H3K79me3	66.01	61.17	58.35	61.03	59.98	59.54	61.05	58.45	56.53	58.00
	H3K9ac	56.79	51.22	45.81	49.35	47.13	46.97	45.12	46.20	44.39	45.40
	H4	80.07	79.26	76.17	78.61	74.05	74.84	76.68	72.09	72.99	71.61
	H4ac	54.19	37.43	33.74	37.14	35.94	36.65	35.09	32.52	33.47	31.80
TFP	0	48.01	44.42	31.04	48.96	52.35	53.33	53.60	46.23	43.79	39.30
(Mouse)	1	81.86	78.94	75.04	81.69	75.91	76.32	75.42	70.14	71.16	70.44
	2	82.98	71.44	61.67	81.71	72.63	75.07	76.83	72.14	73.22	74.75
	3	73.22	44.89	29.17	63.17	65.80	65.29	64.12	62.36	58.19	61.71
	4	46.15	42.48	29.27	42.83	42.76	39.89	40.45	29.79	35.02	32.03
Mean (non-Human)		60.54	51.55	45.31	53.43	51.11	51.41	50.83	48.14	47.56	47.44

Table 6. Performance comparison in terms of Matthews Correlation Coefficient (MCC) metric on the **GUE benchmark datasets that target the ‘Human’ species. For the masking baselines, we pretrain for 1000 steps.** For a fair comparison, we select variants of SoTA models that are trained only on the Human Reference Genome dataset and use the k-mer tokenization scheme (k=6); for DNABert-2, we use the results presented in Table 10 of Appendix of Zhou et al. [36]. The task names are shortened due to space constraints - Promoter Detection (PD), Core Promoter Detection (CPD), Transcription Factor Prediction (TFP) and Splice Site Prediction (Splice). ◇ and ★ indicates that the values are directly taken from Table 6 and 10 of Zhou et al. [36].